

Java internship

Theory

1. Arrays in Java

- **Definition:** A collection of elements of the same type stored in contiguous memory.
- **Declaration:** `int arr[] = {10,20,30,40,50};`
- **Access:** `arr[i]` using index (0-based).
- **Length:** `arr.length` gives the size.

2. Scanner Class

- Used for input from the user.
- Common methods:
 - `nextInt()` → integer input.
 - `nextFloat()` → float input.
 - `nextLine()` → string input (reads entire line).

3. Constructors

- **Default constructor:** No parameters, initializes default values.
- **Parameterized constructor:** Accepts arguments to initialize fields.
- **Called automatically** when an object is created using `new`.

4. Methods

- **Definition:** Block of code performing a task.
- **Return type:** `void` (no return) or specific type (`int`, `String`, etc.).
- **Parameters:** Values passed into methods.

Example:

```
int sum(int a, int b) {  
    return a + b;  
}
```

5. Objects

- **Definition:** Instance of a class.
- **Creation:** Demo obj = new Demo();
- **Access fields/methods:** obj.a, obj.display().

Coding Round Questions

- Arrays → How do you declare and access arrays in Java?
- Scanner → What is the difference between next() and nextLine()?
- Constructors → Explain default vs parameterized constructors.
- Methods → Write a method to calculate sum of two numbers.
- Objects → How do you create and use objects in Java?
- OOP basics → Explain encapsulation, inheritance, and polymorphism briefly.

Machine Coding Round Challenge

Problem: Student Management System

Requirements:

1. Create a Student class with fields: name, age, marks.
2. Add constructors (default and parameterized).
3. Add a method display() to print student details.
4. In main(), create an array of students.
5. Use a loop to print all student details.

```
import java.util.Scanner;

public class Day3 {

    public static void main(String[] args) {

        System.out.println("Arrays");
```

```

int arr[] = {10,20,30,40,50};
// System.out.println(arr[0]);
// System.out.println(arr[2]);
// System.out.println(arr[4]);
// System.out.println(arr[1]);

// for(int i=0; i<arr.length; i++){
//     // System.out.println(i);
//     System.out.println(arr[i]);
// }

Scanner inp = new Scanner(System.in);

// int data = inp.nextInt();
// System.out.println(data);

// float salary = inp.nextFloat();

// System.out.println(salary);

// String name = inp.next();
String name = inp.nextLine();
System.out.println(name);

// 1 byte = > 8 bits
// 4 bytes => 32 bits
// 1 bit => 0 or 1

}
}

```

```

class Demo{

    int a ;    // field

    Demo(){ // default constructor
        a = 100;
        System.out.println("Constructor is called");
    }
    // parameterized constructor
    Demo(int b){
        a = b;
        System.out.println("Parameterized constructor called");
    }
}

```

```

    void display(){ // method
        System.out.println("Display method");
    }

    int sum(int a, int b){ // parameter
        int total = a+b;
        return total;
    }
}

public class ClasssObject {
    public static void main(String[] args) {

        Demo obj = new Demo();
        System.out.println(obj.a);

        Demo obj1 = new Demo(30);
        System.out.println(obj1.a);
        // Demo obj2 = new Demo();
        // Demo obj3 = new Demo();
        // Demo obj4 = new Demo();

        // System.out.println(obj.a);
        // obj.display();

        // int result = obj.sum(10,30); // argument
        // System.out.println(result);

    }
}

```